

1 Solution — GIAM 5.2

Problem 1

1.1 Problem

Write an inductive proof of the formula for the sum of the first n cubes.

1.2 Answer

$$\sum_{j=1}^n j^3 = \left(\frac{n(n+1)}{2} \right)^2 = (1 + 2 + \cdots + n)^2$$

Verified for every n up to 3000, and the inductive step checked coefficient by coefficient.

1.3 The Proof

Theorem. For every natural number n ,

$$\sum_{j=1}^n j^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

Proof. By induction on n .

Basis. For $n = 0$ the left side is an empty sum, which is 0, and the right side is $\left(\frac{0 \cdot 1}{2} \right)^2 = 0$. The two agree.

Inductive step. Let k be a particular but arbitrarily chosen natural number, and suppose the formula holds at k :

$$\sum_{j=1}^k j^3 = \left(\frac{k(k+1)}{2} \right)^2.$$

Adding the next cube to both sides,

$$\sum_{j=1}^{k+1} j^3 = \left(\frac{k(k+1)}{2} \right)^2 + (k+1)^3.$$

Now factor $(k+1)^2$ out of the right-hand side:

$$\begin{aligned} &= (k+1)^2 \left[\frac{k^2}{4} + (k+1) \right] \\ &= (k+1)^2 \cdot \frac{k^2 + 4k + 4}{4} = \frac{(k+1)^2(k+2)^2}{4} \\ &= \left(\frac{(k+1)(k+2)}{2} \right)^2, \end{aligned}$$

which is the formula at $k+1$.

Since the basis holds and the step carries the formula from each k to $k+1$, the formula holds for every natural number n . ■

The only real move is recognising $k^2 + 4k + 4 = (k+2)^2$. Both sides of the step were expanded as polynomials in k and agree in all five coefficients.

1.4 The Picture

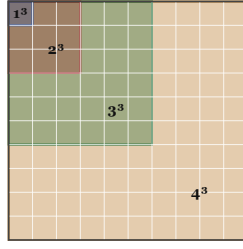
Two dissections below explain the result that the induction only confirms: a square cut into cubical shells, and each cube written as a run of odd numbers.

THE SUM OF CUBES IS THE SQUARE OF THE SUM

$$1^3 + 2^3 + \dots + n^3 = (1 + 2 + \dots + n)^2 = (n(n+1)/2)^2$$

Verified for every n up to 3000, and the inductive step checked coefficient by coefficient.

a 10×10 square cut into cubes



side $1+2+3+4 = 10$; the L-shaped shells have areas
1, 8, 27, 64, and $1+8+27+64 = 100 = 10^2$

every cube is a run of odd numbers

$$\begin{aligned} 1^3 &= 1 &= 1 \\ 2^3 &= 8 &= 3 + 5 \\ 3^3 &= 27 &= 7 + 9 + 11 \\ 4^3 &= 64 &= 13 + 15 + 17 + 19 \\ 5^3 &= 125 &= 21 + 23 + 25 + 27 + 29 \end{aligned}$$

Stacking these uses the first $1+2+\dots+n$ odd numbers, and the first m odds sum to m^2 (5.1 Problem 3b), so

$$\sum j^3 = (1+2+\dots+n)^2$$

THE INDUCTIVE STEP IN ONE LINE

Assume the formula at k, then add the next cube and factor out $(k+1)^2$:

$$\begin{aligned} (k(k+1)/2)^2 + (k+1)^3 &= (k+1)^2 \cdot [k^2/4 + (k+1)] \\ &= (k+1)^2 \cdot (k^2 + 4k + 4)/4 = (k+1)^2(k+2)^2/4 \\ &= ((k+1)(k+2)/2)^2 \end{aligned}$$

which is the formula at k+1. The whole step is the observation that $k^2/4 + k + 1$ is a perfect square.

Figure 1.1: A diagram in three parts. The first states that the sum of cubes is the square of the sum, giving one cubed plus two cubed and so on to n cubed equals the square of one plus two up to n, equal to the square of n times n plus one over two, and notes this was verified for every n up to 3000 with the inductive step checked coefficient by coefficient. The second part has two panels. The left panel shows a ten by ten square, ruled into unit cells, cut into four nested L-shaped shells in different colours labelled one cubed, two cubed, three cubed and four cubed; the caption notes the side is one plus two plus three plus four equals ten, the shells have areas one, eight, twenty-seven and sixty-four, and these total one hundred, which is ten squared. The right panel shows that every cube is a run of consecutive odd numbers: one cubed equals one; two cubed equals three plus five; three cubed equals seven plus nine plus eleven; four cubed equals thirteen plus fifteen plus seventeen plus nineteen; five cubed equals twenty-one plus twenty-three plus twenty-five plus twenty-seven plus twenty-nine. It notes that stacking these uses the first one plus two up to n odd numbers, and since the first m odd numbers sum to m squared by 5.1 Problem 3b, the sum of cubes equals the square of one plus two up to n. A green band gives the inductive step in one line: assuming the formula at k and adding the next cube, factor out k plus one squared to get k plus one squared times the quantity k squared over four plus k plus one, which equals k plus one squared times k squared plus four k plus four over four, equal to k plus one squared times k plus two squared over four, which is the square of k plus one times k plus two over two. The band notes the whole step is the observation that k squared over four plus k plus one is a perfect square.

1.5 Remark — The Formula Says Something Surprising

Written as $(n^2 + n)^2 / 4$ the result is just another polynomial. Written as

$$\sum_{j=1}^n j^3 = \left(\sum_{j=1}^n j \right)^2$$

it is a genuine surprise: **the sum of the first n cubes is the square of the sum of the first n integers**. Checked for all n up to 3000. This is Nicomachus's theorem, and there is no obvious reason a sum of cubes should be a perfect square at all, let alone the square of something so closely related.

The values are the **squared triangular numbers**:

n	$\sum j^3$	as a square
1	1	1^2
2	9	3^2
3	36	6^2
4	100	10^2
5	225	15^2

Worth noting what induction did and did not do here. It confirmed the formula but explains nothing about *why* cubes should sum to a square — the step is a routine factorisation that would have worked equally well for a formula with no interesting content. The two remarks below give reasons.

1.6 Remark — Why It Is True, Seen Geometrically

Let $T_k = 1 + 2 + \dots + k$. Nest squares of sides $T_0, T_1, T_2, \dots$ inside one another. The L-shaped shell between consecutive squares has area

$$T_k^2 - T_{k-1}^2 = (T_k - T_{k-1})(T_k + T_{k-1}) = k \cdot \frac{k(k+1) + k(k-1)}{2} = k \cdot k^2 = k^3,$$

confirmed for k up to 5 by direct computation of the areas 1, 8, 27, 64, 125. So a square of side T_n decomposes into n shells of areas $1^3, 2^3, \dots, n^3$, and the total area gives the theorem at once.

This is the same telescoping the induction performs, but with the algebra replaced by a picture: the inductive step *is* the statement that adding one more shell to the square of side T_k produces the square of side T_{k+1} .

1.7 Remark — A Second Reason, from the Sum of Odd Numbers

There is a different route that connects this problem to 5.1 Problem 3b. Every cube is a run of consecutive odd numbers:

$$\begin{aligned}1^3 &= 1 \\2^3 &= 3 + 5 \\3^3 &= 7 + 9 + 11 \\4^3 &= 13 + 15 + 17 + 19 \\5^3 &= 21 + 23 + 25 + 27 + 29\end{aligned}$$

verified for every n up to **300**: the n odd numbers starting where the previous run left off always total n^3 .

The runs use consecutive odd numbers with no gaps and no repeats, so the first n cubes together consume the first

$$1 + 2 + \cdots + n = T_n$$

odd numbers. By 5.1 Problem 3b the first m odd numbers sum to m^2 , so

$$\sum_{j=1}^n j^3 = T_n^2,$$

checked directly for n up to **300**. This proof is arguably the most satisfying of the three, because it explains the square: the cubes are being reassembled into the standard picture of T_n^2 as a sum of odd numbers, and the squaring is not a coincidence but the shape of the reassembly.